

Kavya Patel

602-566-0736 kavyarp197@gmail.com linkedin.com/in/kavya-patel19

Summary

Skilled software engineer with strong problem-solving abilities and experience in designing, developing, and maintaining scalable web and software applications. Proficient in modern programming languages, frameworks, and best practices. Passionate about building efficient, user-friendly solutions and collaborating in agile teams to deliver high-quality products that meet business and user needs.

Experience

Software Development Intern, Argusoft Ind Ltd.

January 2023 – May 2023

- Developed a full-stack web application using Node.js, Vue.js, and PostgreSQL to automate unit billing calculations, with support for custom invoice generation, recurring billing, and dynamic templates.
- Implemented real-time data validation and error-handling mechanisms, improving billing accuracy and reducing processing errors by 20%.
- Scaled the platform to handle 10x more users by migrating to AWS, optimizing database queries, and integrating CI/CD pipelines, leading to a 50% increase in deployment frequency and 30% reduction in downtime.
- Utilized a modern tech stack including JavaScript, Node.js, Vue.js, Fabric.js, PostgreSQL, and AWS to build and maintain a secure, responsive, and efficient billing system.

Assistant Researcher, Undergraduate Research Program

August 2022 – December 2022

- Developed a real-time vehicle number plate verification system using Python and OpenCV, achieving 87% detection accuracy under diverse lighting conditions.
- Processed and validated a dataset of 5,000+ images, reducing recognition time from 1.2s to under 500ms per frame through optimized image handling.
- Enhanced performance by resolving preprocessing bottlenecks, boosting detection speed and reliability.
- Applied contour detection and morphological operations to improve plate localization and character segmentation.

Projects

Online Learning Platform | Node.js, MongoDB, Angular, Jenkins

Fall 2024

- Designed and developed a scalable online learning platform using Angular for the front-end and Node.js with Express for the back-end, serving 500,000+ users.
- Implemented RESTful APIs to manage course content, user authentication, and payment processing, reducing API response times by 25% through efficient database querying with MongoDB.
- Led the integration of real-time chat functionality using Socket.io, enhancing user engagement by 20% through interactive course discussions and live Q&A sessions.

FitGenie | Python, Node.js, React.js, MySQL

Spring 2023

- Developed a fitness recommendation system for 150+ users, generating personalized workout and nutrition plans that improved user retention by 30%.
- Built a secure Node.js backend with REST APIs and MySQL integration to manage 10,000+ user data points, and created a scheduling system with trainer availability, reducing consultation wait times by 40%.
- Designed a modular, mobile-first React interface enabling smooth navigation across dashboards, booking, and progress tracking for enhanced user experience.

Warehouse Management System | Java, Springboot, AWS, Docker

Fall 2022

- Built a WMS that reduced order processing time by 35% and improved shipment accuracy by 25% through automated inventory validation and order tracking.
- Deployed on AWS EC2 using Docker containers, maintaining 99.8% uptime across 3 months with support for auto-scaling under high traffic.
- Integrated AWS CloudWatch alerting and logging systems to enhance monitoring, enabling 50% faster issue resolution during peak hours.

Education

Masters of Science in Computer Science

May 2025

Arizona State University, Arizona, USA

3.5/4.0

Bachelor of Technology in Computer Science And Engineering

May 2023

Ahmedabad University, Gujarat, India

3.4/4.0

Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS

Frameworks & Technologies: React.js, Vue.js, Angular, Node.js, Express.js, Django, Flask, FastAPI, TensorFlow, PyTorch, Keras, OpenCV, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn

Databases and Cloud Technologies: MySQL, PostgreSQL, MongoDB, Firebase, SQLite, AWS, Microsoft Azure, Docker